GEOMETRY QUEST · WORLD 16

Fractals

ANSWER SHEET · for the grown-up

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

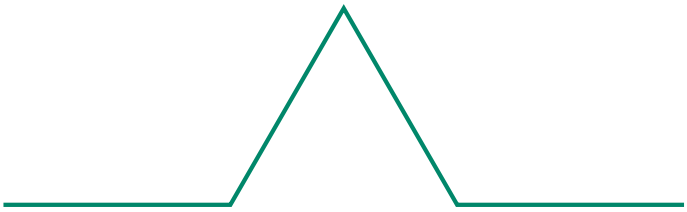
1 Cantor set: each step removes the middle third of every piece. Complete the table.

step	pieces	length left (start = 1)
0	**1**	**1**
1	**2**	**2/3**
2	**4**	**4/9**
3	**8**	**8/27**
4	**16**	**16/81**

2 Koch snowflake: start with a triangle of side 9 cm. Each step, every edge becomes 4 edges, each a third as long. Complete the table.

step	edges	perimeter
0	**3**	**27 cm**
1	**12**	**36 cm**
2	**48**	**48 cm**
3	**192**	**64 cm**

3 Draw step 1 of the Koch curve on this 9 cm line: split it into three 3 cm parts, draw an equilateral triangle on the middle part, then rub out its base.



4 Fractal hunt: find three things in nature or at home that repeat their pattern when you zoom in. Draw one of them.

e.g. fern leaf, broccoli/cauliflower, tree branches, river deltas, frost, lightning.

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0